Radiation Incident

TB-07 | Trauma & Burns

Quick flow

01 Radiation Hazard [EMT / AEMT / Paramedic]

Stage until Incident Command identifies a safe zone and the appropriate PPE and radiation monitoring/dosimetry when required.

02 Life Threat? [EMT / AEMT / Paramedic]

Do not delay lifesaving airway, breathing, hemorrhage care, or transport for decontamination.

03 Contaminated? [EMT / AEMT / Paramedic]

If radioactive material is on clothing or skin, carefully remove outer clothing, bag and label belongings, then gently wash with soap and lukewarm water. Do not decontaminate an exposure-only patient.

04 Assess Exposure [EMT / AEMT / Paramedic]

Source • time • distance • shielding • symptoms • associated blast/burn/trauma • survey or dose information provided by radiation authorities.

05 Coordinate Experts [AEMT / Paramedic]

Incident Command/HazMat • local radiation safety • receiving facility • Medical Control.

06 Controlled Transport [EMT / AEMT / Paramedic]

Notify the destination early. If unstable, transport promptly with contamination precautions rather than delaying for completed decontamination.

Overview

- Radiation exposure, external contamination, and internal contamination are distinct problems; exposure alone does not make the patient or ambulance contaminated.
- Treat life threats first while using time, distance, shielding, PPE, and contamination control.

Provider-Level Actions

Perform all immediately indicated care within scope without delaying rapid trauma transport.

EMT

- Use appropriate PPE and contamination control, treat immediate threats, and prevent spread. Exposure without contamination does not require patient or ambulance decontamination.
- For known or suspected external contamination, carefully remove outer clothing, bag and label belongings, and gently wash exposed skin or hair with soap and lukewarm water; do not scrub or injure the skin.
- Record source information, exposure time, distance, shielding, symptoms, decontamination performed, and any survey or dose information provided by radiation authorities.

AEMT

- Perform all EMT care plus vascular access and supportive treatment within standing orders.
- Coordinate monitoring, ambulance preparation, and transport precautions with Incident Command.

Paramedic

- Perform all prior care plus advanced supportive care and consultation with Medical Control, HazMat, local radiation safety, and the receiving facility.
- Manage associated trauma, burns, airway injury, shock, or dysrhythmia under the appropriate protocols. Do not give potassium iodide or other radiation countermeasures unless specifically directed by public-health or radiation authorities.

Indications

- Patient with suspected radiation incident requiring prehospital assessment or treatment.

Warnings / precautions

- Do not enter a radiation hot zone without authorization, training, appropriate PPE, and required monitoring/dosimetry.
- Contamination alone is not a reason to withhold lifesaving care or delay transport of a seriously injured patient.

Source and references

Current Tennessee EMS scope of practice, verified individual credentials, and Claiborne County standing orders control provider authorization.

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